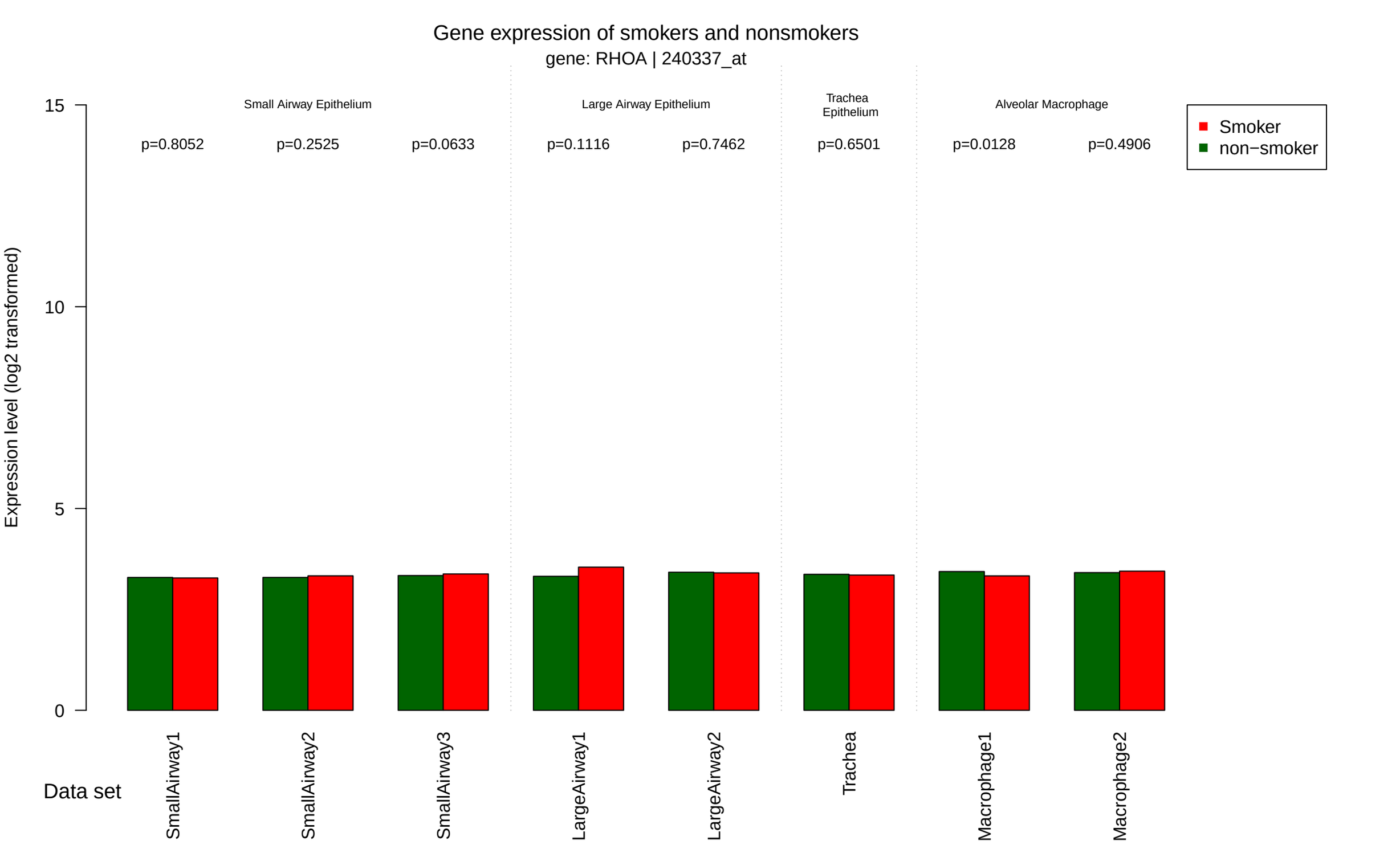
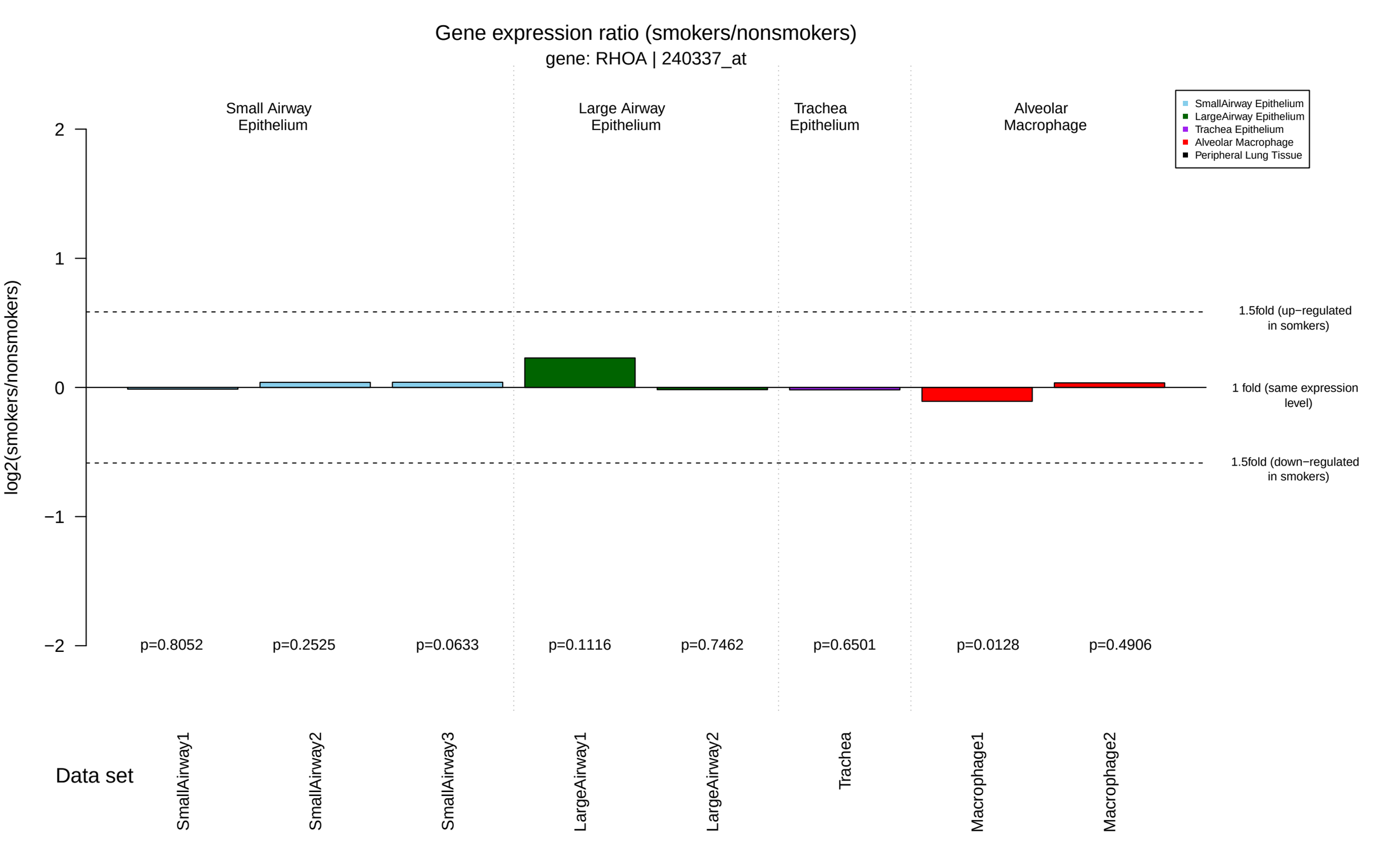
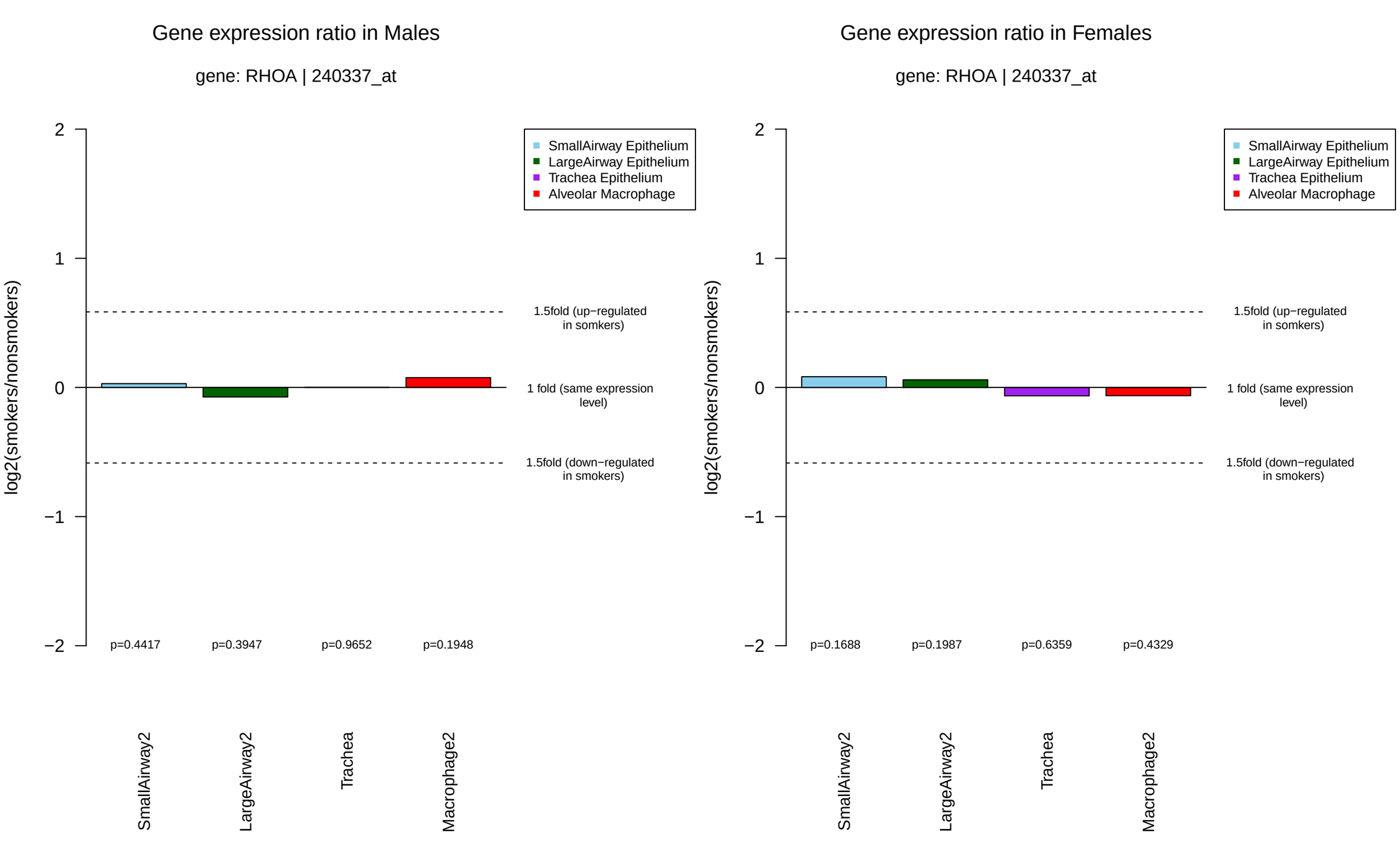
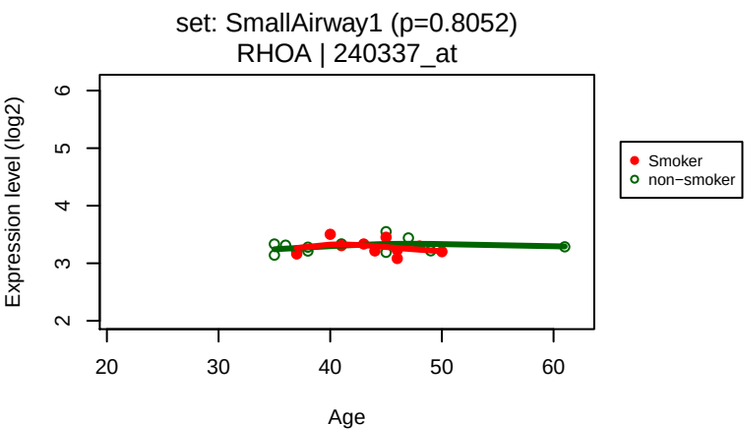
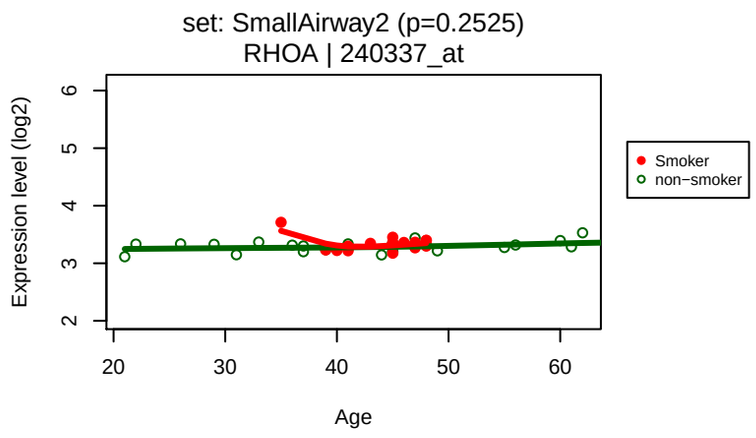
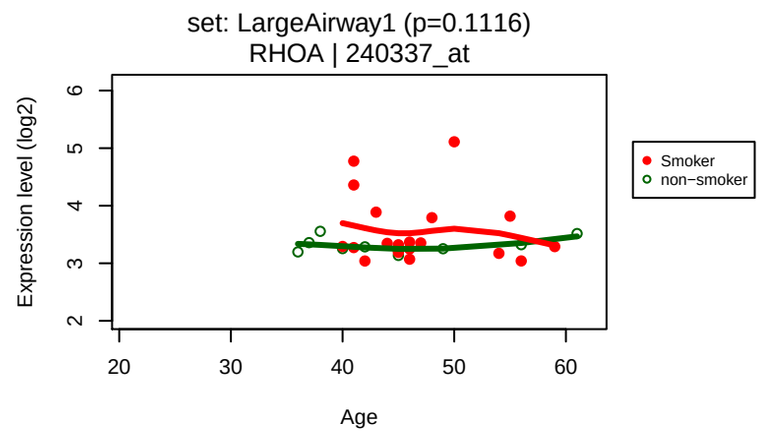

Gene expression of smokers and nonsmokers
gene: RHOA | 240337_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.8052
p=0.2525
p=0.0633
p=0.1116
p=0.7462
p=0.6501
p=0.0128
p=0.4906
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

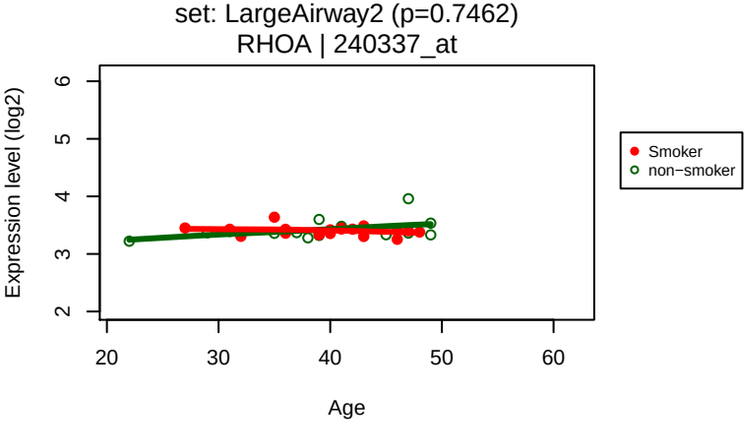

Gene expression ratio (smokers/nonsmokers)
gene: RHOA | 240337_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8052
p=0.2525
p=0.0633
p=0.1116
p=0.7462
p=0.6501
p=0.0128
p=0.4906
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

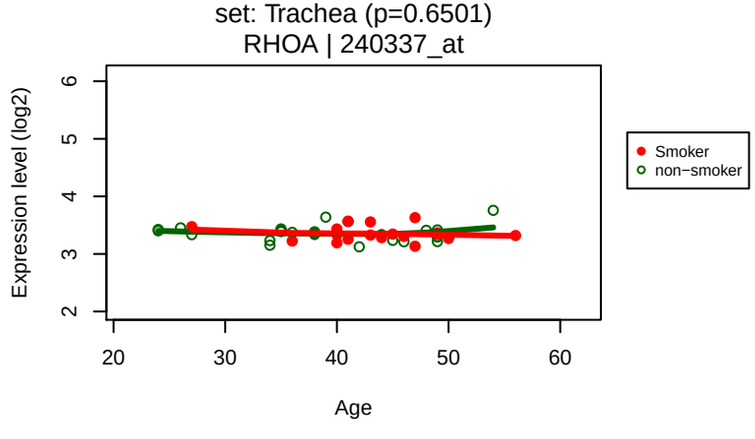

Gene expression ratio in Males
gene: RHOA | 240337_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4417
p=0.3947
p=0.9652
p=0.1948
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: RHOA | 240337_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1688
p=0.1987
p=0.6359
p=0.4329
SmallAirway2
LargeAirway2
Trachea
Macrophage2

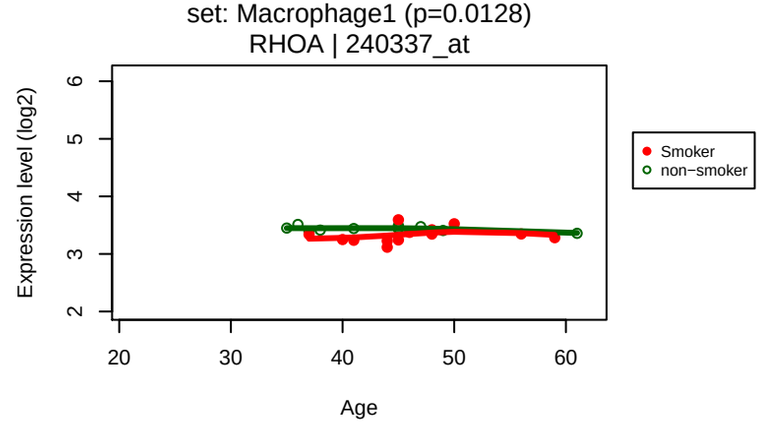

set: SmallAirway1 (p=0.8052)
RHOA | 240337_at
Expression level (log2)
Age
Smoker
non−smoker

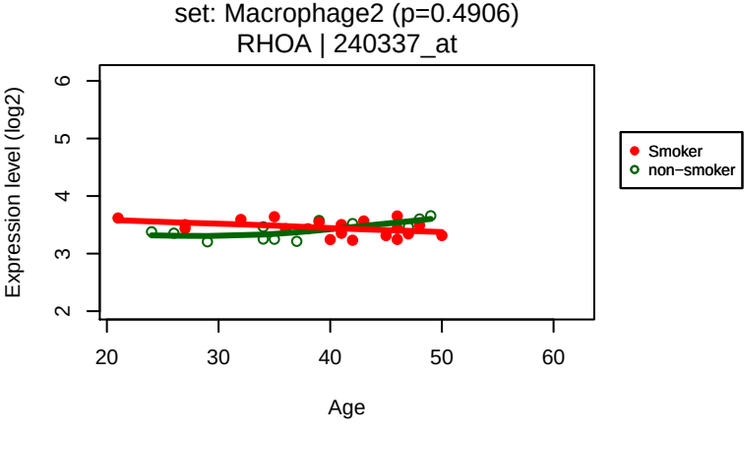

set: SmallAirway2 (p=0.2525)
RHOA | 240337_at
Expression level (log2)
Age
Smoker
non−smoker

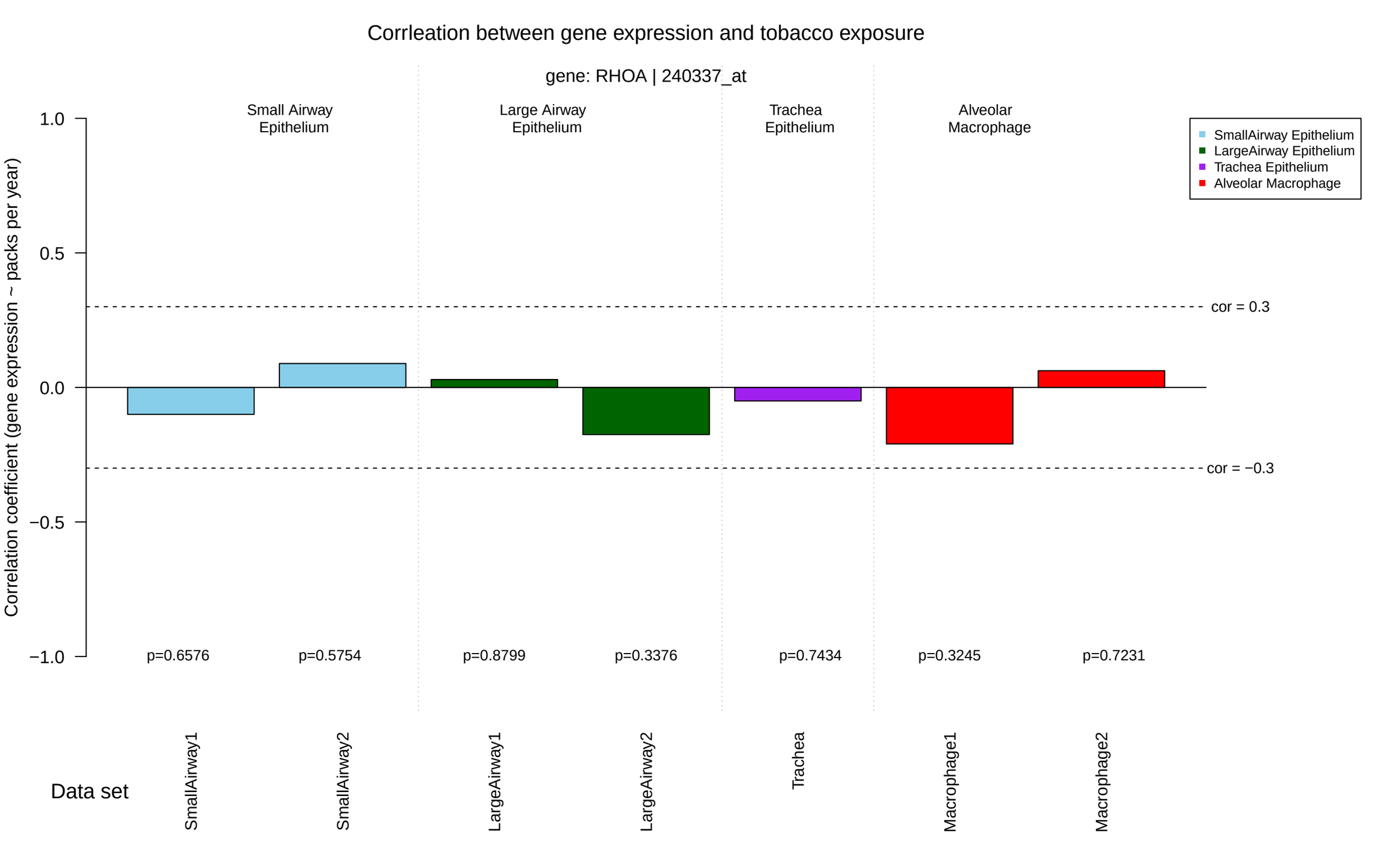

set: LargeAirway1 (p=0.1116)
RHOA | 240337_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.7462)
RHOA | 240337_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.6501)
RHOA | 240337_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0128)
RHOA | 240337_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.4906)
RHOA | 240337_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: RHOA | 240337_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.6576
p=0.5754
p=0.8799
p=0.3376
p=0.7434
p=0.3245
p=0.7231
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2